

Predictive Modeling in Diabetes

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Sept. 20, 2024

Adverse outcomes of diabetes

- ▶ Diabetes is an increasingly prevalent disease (13% of adults in U.S.)
- ▶ Chronic complications incl. retinopathy, kidney disease, & CVD
- ▶ Lowering blood glucose levels decreases the risk of complications
- ▶ **Goal:** identify diabetes patients at high risk of treatment failure
 - $\text{HbA1c} \geq 7.0\%$ (i.e., avg. blood glucose ≥ 150 mg/dL) over 12 mos.

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Target population data



Predictive Model

$\text{HbA1c} < 7.0\%$

$\text{HbA1c} \geq 7.0\%$

Low Risk

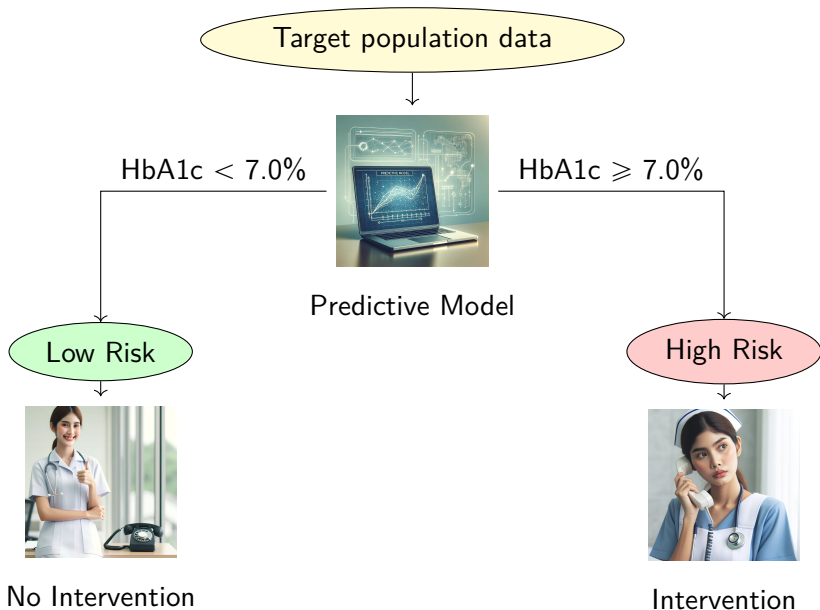
High Risk



No Intervention



Intervention



Target population data

$HbA1c < 7.0\%$

$HbA1c \geq 7.0\%$

Low Risk

High Risk

- ▶ Patients at high risk can be provided with additional resources (e.g., phone calls from nurses; more frequent appointments & testing).
- ▶ These resources are costly and cannot be provided to everyone.



No Intervention



Intervention

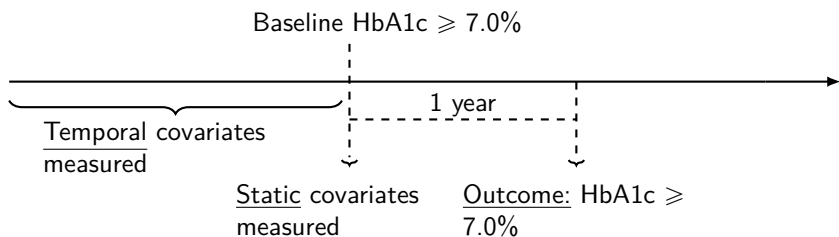
Study	TransformEHR (Yang et al., 2023)
Model	Encoder-decoder transformer + cross-attention
Tasks	Disease/outcome agnostic prediction; single disease/outcome prediction
Population	VHA EHRs (6.4M patients)
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Advantages	Pretraining obj. enables model to predict all diseases/outcomes of a visit	Handles EHR narrative and free-form text
Disadvantages	Limited to ICD10 codes	Overfitting risk in less diverse datasets

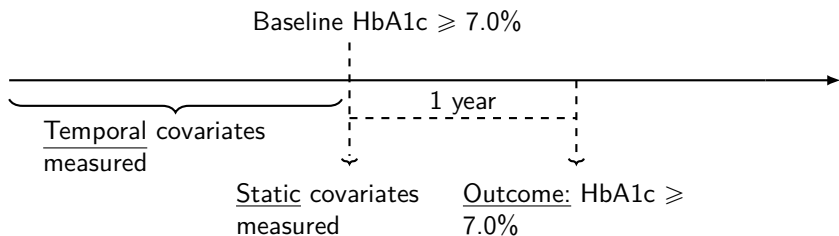
Study	TransformEHR (Yang et al., 2023)	Foresight (Kraljevic et al., 2024)	TransEHR (Xu et al., 2023)
Model	Encoder-decoder transformer + cross-attention	Built on top of GPT-2	Self-supervised transformer
Tasks	Disease/outcome agnostic prediction; single disease/outcome prediction	Adverse clinical outcomes prediction	In-hospital mortality; length-of-stay; acute care phenotypes
Population	VHA EHRs (6.4M patients)	UK hospitals (811K patients)	ICU patients (PhysioNet, ...)
Advantages	Pretraining obj. enables model to predict all diseases/outcomes of a visit	Handles EHR narrative and free-form text	Effective with limited labeled data
Disadvantages	Limited to ICD10 codes	Overfitting risk in less diverse datasets	Limited to ICU patients from public datasets

Study	TransformEHR (Yang et al., 2023)	Foresight (Kraljevic et al., 2024)	TransEHR (Xu et al., 2023)	BERT/XLNet (Antikainen et al., 2023)
Model	Encoder-decoder transformer + cross-attention	Built on top of GPT-2	Self-supervised transformer	BERT and XLNet (autoregressive transformer)
Tasks	Disease/outcome agnostic prediction; single disease/outcome prediction	Adverse clinical outcomes prediction	In-hospital mortality; length-of-stay; acute care phenotypes	Mortality risk prediction over 6 months
Population	VHA EHRs (6.4M patients)	UK hospitals (811K patients)	ICU patients (PhysioNet, ...)	Finnish CVD patients (23K)
Advantages	Pretraining obj. enables model to predict all diseases/outcomes of a visit	Handles EHR narrative and free-form text	Effective with limited labeled data	XLNet is better capable of capturing long-term dependencies
Disadvantages	Limited to ICD10 codes	Overfitting risk in less diverse datasets	Limited to ICU patients from public datasets	Focused on single prediction task

Diabetes treatment risk among primary care patients



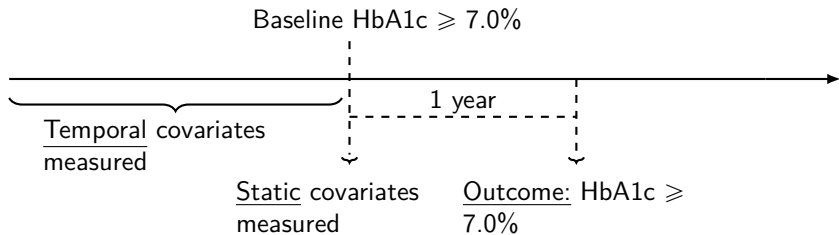
Diabetes treatment risk among primary care patients



Temporal characteristics:

- Diagnoses (ICD-10 codes)
- Procedures (CPT codes)
- Labs (codes & results)
- Medications
- Text (e.g., clinician notes)
- Vitals

Diabetes treatment risk among primary care patients



Temporal characteristics:

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Static characteristics:

- Baseline HbA1c value
- Age
- Gender
- Marital status
- Health insurance status
- Primary language
- SDI based on zip code
- Veteran status

Outpatient data from MGB EHRs

- ▶ Diabetes patients (18-75) with baseline HbA1c ≥ 7.0 during study period (2000 – 2022) and an A1c measurement 12 mos. after
- ▶ Temporal data collected beginning 1990
- ▶ 55.7K patients, with 10.7M events and 164.2M measurements
 - Patient-level train/val/test split: 38960 (70%); 11131 (20%); 5566 (10%)
- ▶ # unique codes: 33,803
 - Diagnoses (ICD-10): 19,095
 - Non-lab procedures (CPT): 9,056
 - Labs: 5,652

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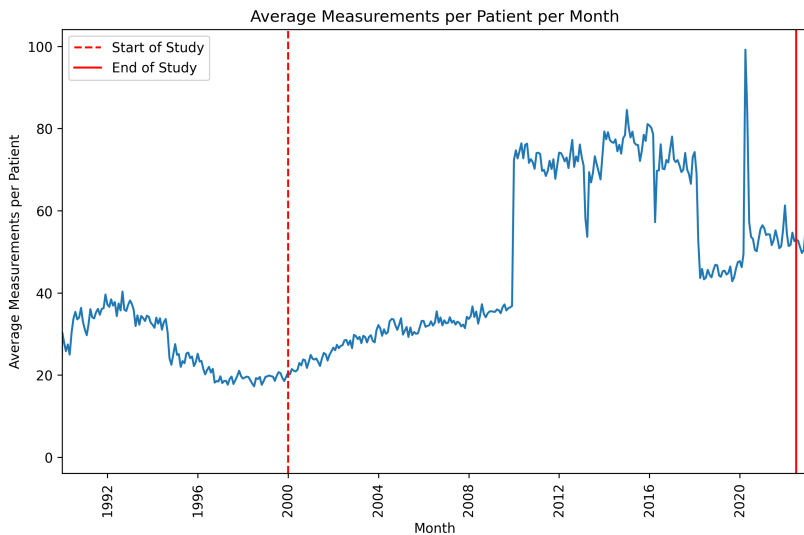
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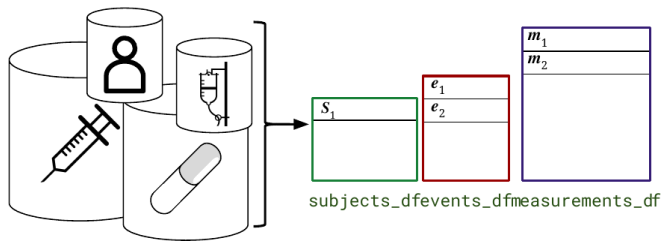
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Summary statistics: static covariates & outcome

Covariate	Mean	Std. Dev.	Min	Max
Age	56.69	11.98	18.0	75.0
English language	0.84	0.37	0.24	1.0
Female	0.46	0.50	0.0	1.0
Baseline HbA1c (%)	8.49	1.80	7.0	19.9
Married	0.53	0.50	0.0	1.0
Medicare/Medicaid	0.58	0.49	0.0	1.0
SDI score	48.06	32.44	0	100
Veteran	0.07	0.26	0.0	1.0
Outcome: HbA1c $\geq$ 7%	0.55	0.50	0.0	1.0



“Event stream” dataset (McDermott et al., 2024)

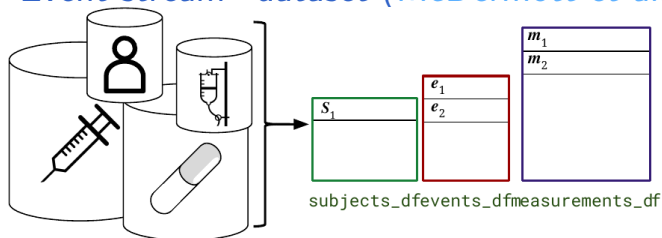


`subjects_df` for static data (55.7K)

`events_df` containing event timestamps & types per subject (10.8M)

`dynamic_measurements_df` storing all observed measurements (164.1M)

"Event stream" dataset (McDermott et al., 2024)



Example rows of subjects_df.

subject_id	Female	Married	GovIns	English	Veteran	InitialA1c	AgeYears	SDI_score	static_indices	static_ measurement_indices
55397	1	0	0	1	0	10.6	37	77	[2, 3, ..., 113]	[1, 3, ..., 80]
29939	0	1	0	1	0	7.6	38	5	[1, 4, ..., 83]	[1, 3, ..., 80]
54566	1	0	1	1	0	8.4	59	5	[2, 3, ..., 83]	[1, 3, ..., 80]
52437	0	0	0	1	0	7.0	61	34	[1, 3, ..., 98]	[1, 3, ..., 80]

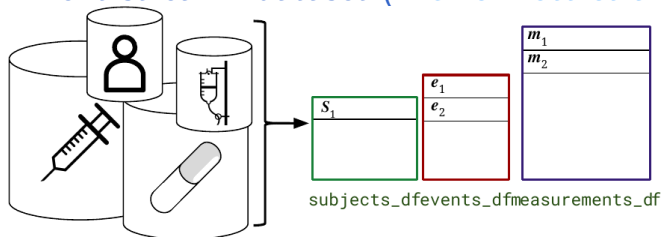
static_measurement_indices vocab:

{'Female': 1, 'Married': 3, 'GovIns': 5, 'English': 7, 'Veteran': 9, 'InitialA1c': 11, 'AgeYears': 42, 'SDI_score': 80}

static_indices vocab:

{ 'Female': {'0': 1, '1': 2}, 'Married': {'0': 3, '1': 4}, ..., 'InitialA1c': {'7-7.10': 11, ...},
'AgeYears': {'18-27': 42, ...}, 'SDI_score': {'0-1': 80, ...}}

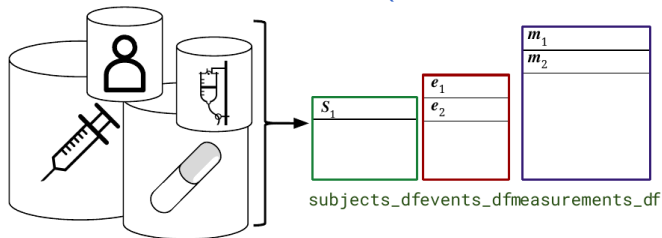
"Event stream" dataset (McDermott et al., 2024)



First and last 4 rows of events_df for subject_id=52437.

event_id	subject_id	timestamp	event_type
10186535	52437	1997-01-31 00:00:00	DIAGNOSIS
10186536	52437	1997-03-14 00:00:00	DIAGNOSIS
10186537	52437	1997-04-25 00:00:00	DIAGNOSIS
10186538	52437	1997-05-08 00:00:00	DIAGNOSIS
.....			
10186647	52437	2022-03-04 00:00:00	LAB
10186648	52437	2022-03-07 00:00:00	LAB
10186649	52437	2022-03-09 00:00:00	LAB
10186650	52437	2022-09-06 00:00:00	LAB

“Event stream” dataset (McDermott et al., 2024)



First and last 4 rows of `measurements_df` for `subject_id=52437`.

measurement_id	event_id	subject_id	timestamp	dynamic_indices	dynamic_values
154339500	10186535	52437	1997-01-31 00:00:00	26332	null
154339501	10186535	52437	1997-01-31 00:00:00	26410	null
154339502	10186536	52437	1997-03-14 00:00:00	26003	null
154339503	10186536	52437	1997-03-14 00:00:00	26326	null
.....					
154340443	10186650	52437	2022-09-06 00:00:00	17636	60.0
154340444	10186650	52437	2022-09-06 00:00:00	15636	82.0
154340445	10186650	52437	2022-09-06 00:00:00	23024	143.0
154340446	10186650	52437	2022-09-06 00:00:00	12415	200.0

Example DL representation

subject_id	time	dynamic_indices	dynamic_measurement_indices	dynamic_values	static_indices	static_measurement_indices
52437	[0, 0, ...9349]	[26332, 26410, ..., 12415]	[1, 1, ..., 3]	[null, null, ..., 200.0]	[1, 3, ...98]	[1, 3, ..., 80]

time: days from start of a patient's record (e.g., ≈ 25 years);
length of array = number of events a subject has.

Example DL representation

subject_id	time	dynamic_indices	dynamic_measurement_indices	dynamic_values	static_indices	static_measurement_indices
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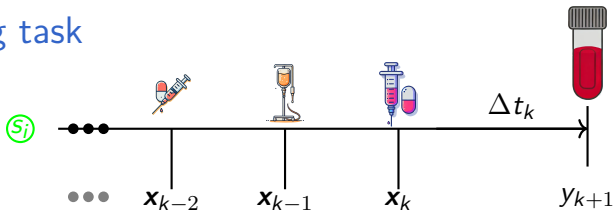
dynamic_indices_vocab sample:

{ '26332': 'R53.83_ICD10', '26410': 'R78.81_ICD10', ..., '12415': 'CHOLESTEROL' }

dynamic_measurement_indices_vocab:

{ 'DIAGNOSIS': 1, 'PROCEDURE': 2, 'LAB': 3 }

Modeling task



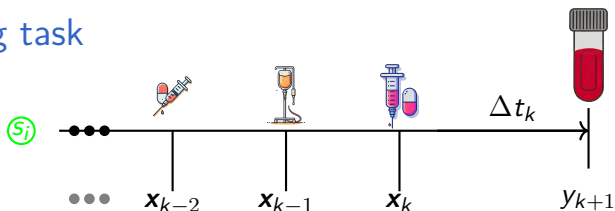
- ▶ For a single subject s_i , their temporal data consist of a sequence of events $\mathbf{x}_k$ for $k = 1, 2, \dots, N$ occurring at times t_k
- ▶ Each event is a collection of covariate measurements recorded at time t_k
- ▶ Model the prob. distr. for a patient's label given their history $\mathbf{h}_{i-1}$:

$$p(y_{k+1} \mid \underbrace{\mathbf{x}_{k-2}, \mathbf{x}_{k-1}, \mathbf{x}_k}_{\mathbf{h}_{i-1}})$$

- ▶ Define a model parameterized by θ that approximates this probability:

$$f_{\theta}(y_{k+1}, \mathbf{h}_{i-1}) = p(y_{k+1} \mid \mathbf{h}_{i-1})$$

Modeling task



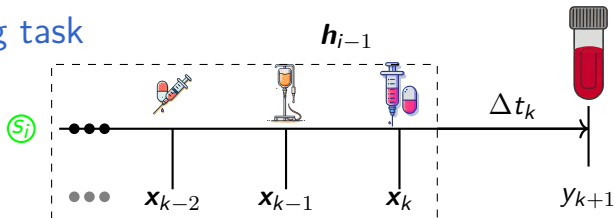
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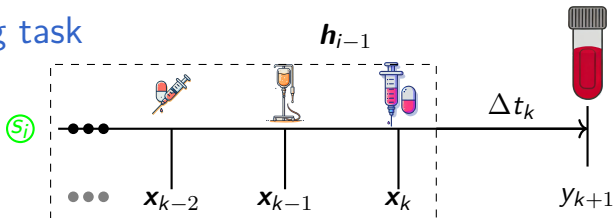
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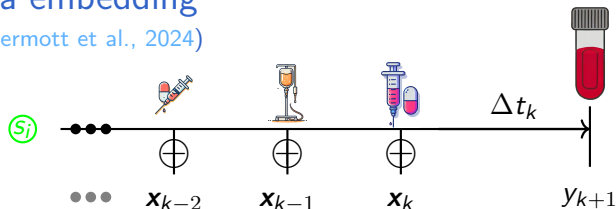
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Data embedding

(McDermott et al., 2024)

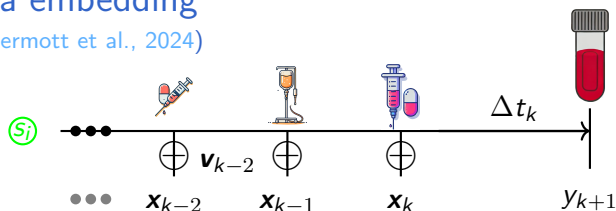


Convert properties of patient's events into fixed-sized embedding vectors using a numerical and categorical embedding bag approach.

- ▶ Assign a vector of normalized numerical features $\mathbf{v}_{k-2}$ for event x_{k-2}
- ▶ Take all properties (e.g., codes) from the event and arrange them into a single unified vocabulary.
- ▶ Develop an embedding layer that associates each categorical code with an embedding vector $\mathbf{e}_i$.

Data embedding

(McDermott et al., 2024)

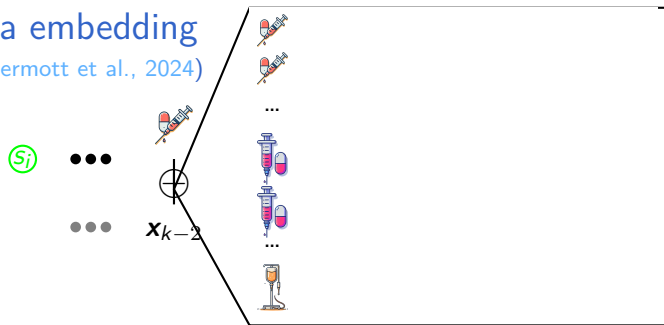


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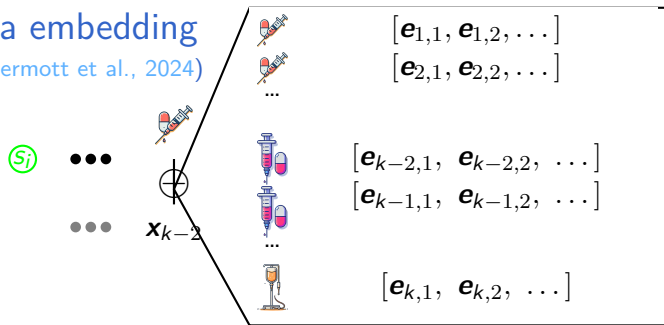


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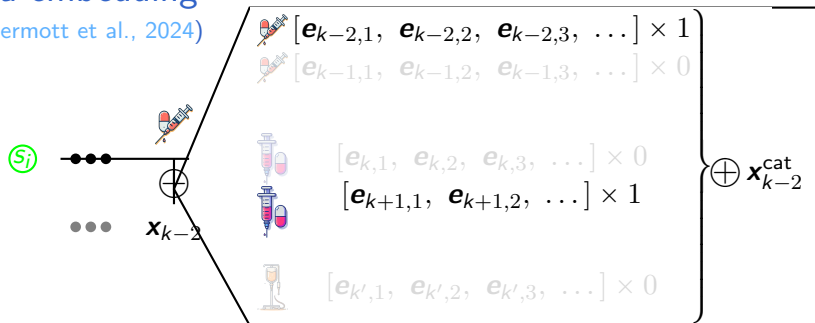


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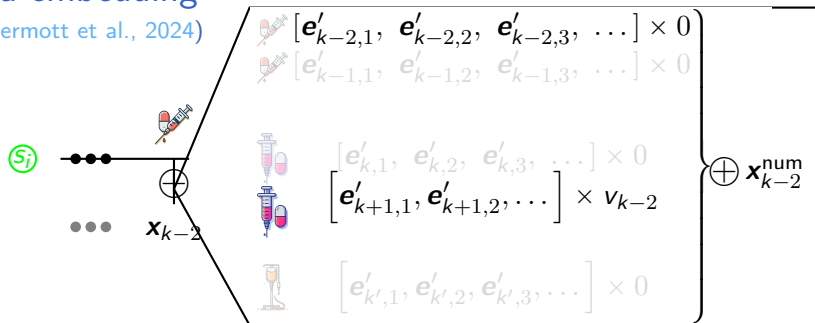


To perform aggregation step using split embedding scheme:

1. Cat. embedding stage: zero out properties not observed within the event; assign 1 to obs. cat. properties; take weighted sum over all observed cat. embeddings
2. Num. embedding stage: zero out observed properties without value; assign v_{k-2} to num. properties
3. Combine embeddings to form final fixed-size embedding

Data embedding

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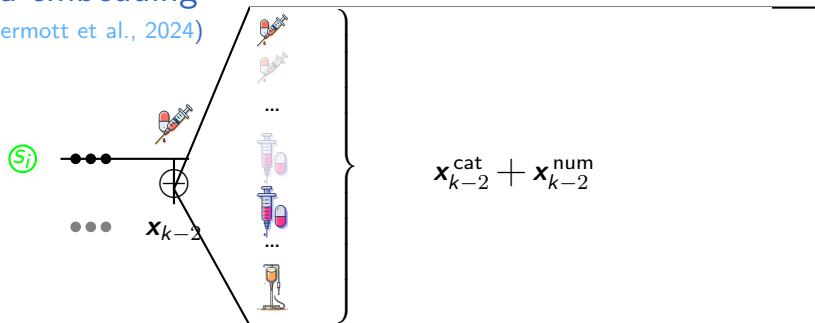


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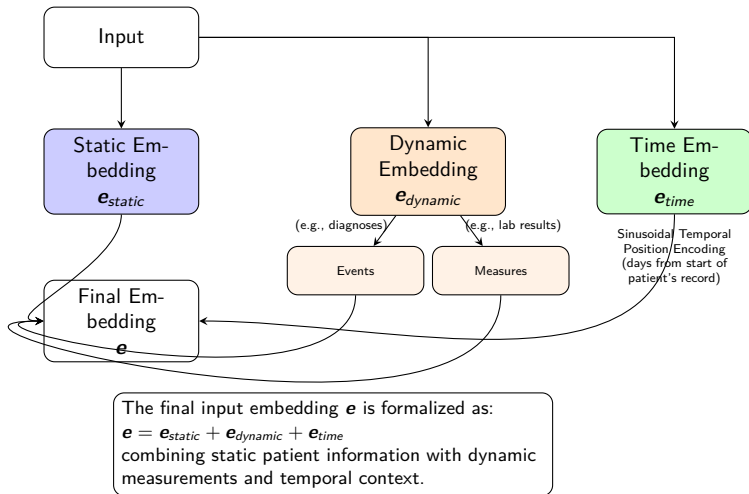
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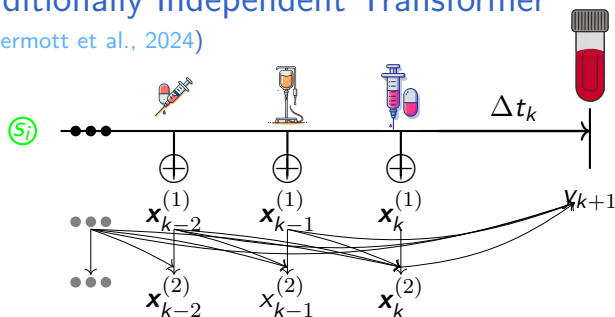
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Multi-level Embedding Scheme



Conditionally Independent Transformer

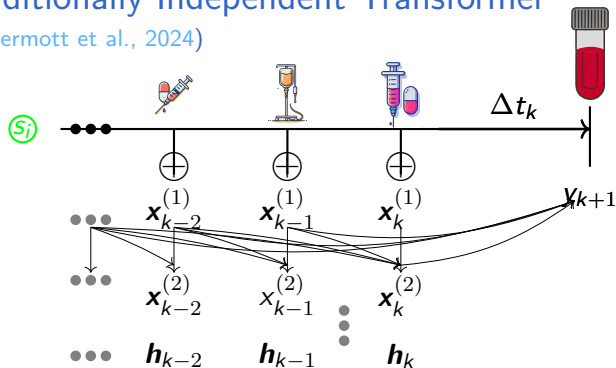
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1. Process embeddings via transformer with multi-headed self-attention
2. Pass embeddings through multiple layers to produce final hidden state h_k
3. Predict label y_{k+1} given h_k
4. Minimize the binary cross-entropy loss function (BCEWithLogitsLoss)

Conditionally Independent Transformer

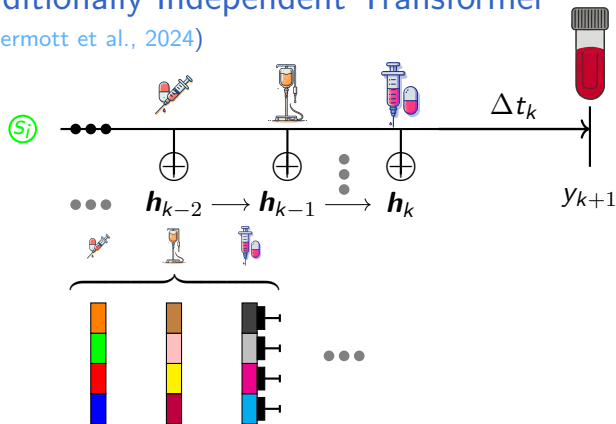
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Conditionally Independent Transformer

(McDermott et al., 2024)

Context probability:

$$p \left(y_{k+1} \mid \underbrace{\mathbf{x}_1, \dots, \mathbf{x}_k}_{\mathbf{h}_k} \right)$$

Model prediction:

$$\hat{y}_i = \sigma (f_{\theta}(\mathbf{h}_k))$$

Loss fn.:

$$\mathcal{L} = - [y_{k+1} \log(\hat{y}_{k+1}) + (1 - y_{k+1}) \log(1 - \hat{y}_{k+1})]$$

1. Process embeddings via transformer with multi-headed self-attention
2. Pass embeddings through multiple layers to produce final hidden state $\mathbf{h}_k$
3. Predict label y_{k+1} given $\mathbf{h}_k$
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Tabular vs. temporal transformers

Model	Vocab size	Labs?	Discr. num. features?	# attn. heads	Head dim.	Depth	Hidden dim.	# trainable params.	AUROC		F1	
									Val.	Test	Val.	Test
Tabular												
FT-Transf.	–	N	N	16	16	3	128	806.7K	0.66	–	–	–
ResNet-like	–	N	N	-	-	3	128	266.8K	0.63	–	–	–
Temporal												
Transformer	33804	Y	Y	8	64	4	256	41.8M	0.90	0.90	0.87	0.87
									$\pm 4 \times 10^{-4}$		± 0.001	
Transformer	3238	Y	N	8	64	4	512	7.4M	0.74	0.73	–	–
Transformer	1630	N	N	8	64	6	512	8.5M	0.68	0.68	–	–

FT-Transformer Model: adaptation of the transformer architecture for tabular data, converting features to embeddings and applying a stack of transformer layers to these embeddings (Gorishniy et al., 2021). [More info.](#)

Tabular vs. temporal transformers

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									Val.	Test	Val.	Test
Tabular												
FT-Transf.	–	N	N	16	16	3	128	806.7K	0.66	–	–	–
ResNet-like	–	N	N	-	-	3	128	266.8K	0.63	–	–	–
Temporal												
Transformer	33804	Y	Y	8	64	4	256	41.8M	0.90	0.90	0.87	0.87
									$\pm 4 \times 10^{-4}$		± 0.001	
Transformer	3238	Y	N	8	64	4	512	7.4M	0.74	0.73	–	–
Transformer	1630	N	N	8	64	6	512	8.5M	0.68	0.68	–	–

ResNet-like Baseline: uses residual networks adapted for tabular data ([Gorishniy et al., 2021](#)). [More info.](#)

Tabular vs. temporal transformers

Model	Vocab size	Labs?	Discr. num. features?	# attn. heads	Head dim.	Depth	Hidden dim.	# trainable params.	AUROC		F1	
									Val.	Test	Val.	Test
Tabular												
FT-Transf.	–	N	N	16	16	3	128	806.7K	0.66	–	–	–
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Conditionally Independent Transformer: Each feature is processed independently into fixed-sized embedding vectors. These embeddings are then combined using an attention mechanism, allowing the model to capture dependencies and interactions over temporal data ([McDermott et al., 2024](#)). [More info.](#)

Next steps

1. Integrate additional patient data such as vital signs, medications, and clinician notes
2. Use attention mechanisms to identify temporal patterns predictive of diabetes treatment failure
 - Apply attention mechanism for static covariates
 - (Integrated Gradients for feature attribution)
3. More advanced transformer architectures
 - e.g., one that doesn't assume intra-event covariates are conditionally independent from each other given the patient's history
4. Effect of historical EHR length on performance
 - transformers benefit from EHRs with longer histories (Yang et al., 2023)

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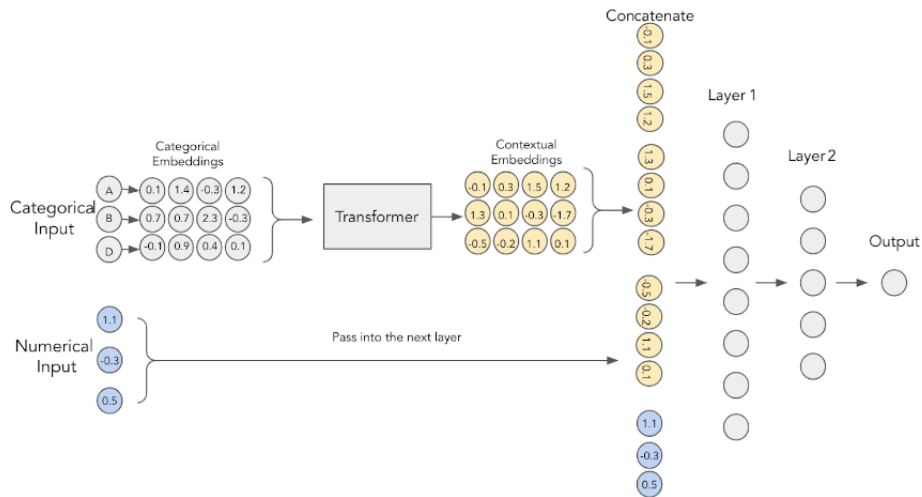
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Questions?

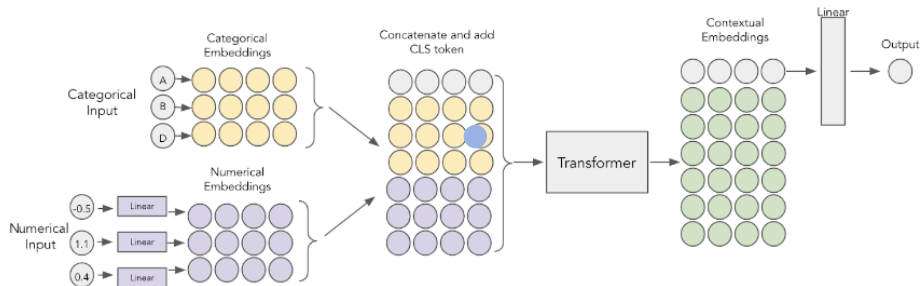
Tab Transformer

Huang et al. (2020)



FT-Transformer (Feature Tokenizer Transformer)

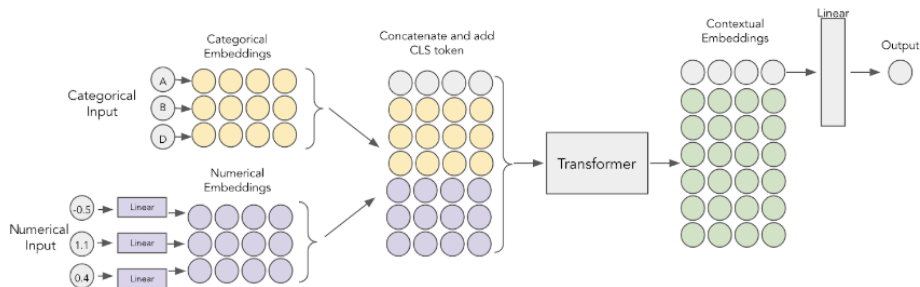
Gorishniy et al. (2021)



CLS token embeddings used as input to final layer, whereas TabTransformer concatenates all the embeddings

FT-Transformer (Feature Tokenizer Transformer)

Gorishniy et al. (2021)



CLS token embeddings used as input to final layer, whereas TabTransformer concatenates all the embeddings

Numerical embeddings pass through Transformer together with the categorical ones; TabTransformer contextualizes only categorical embeddings

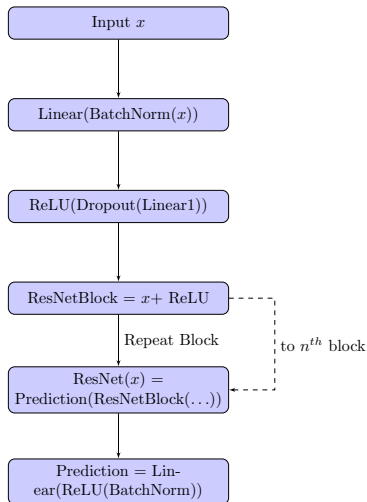
Tab Transformer

ResNet-like baseline

Klambauer et al. (2017); Gorishniy et al. (2021)

Residual networks adapted to FNNs using residual blocks with n layers

Residual block is simplified compared to the original architecture (He et al., 2016)



Hyperparameter optimization

- ▶ Use Ray Tune ASHA tuning algorithm ([Liaw et al., 2018](#))
- ▶ Randomly sample 50 hyperparameter configs.
- ▶ Evaluate configs. on max. validation AUROC
- ▶ Run each config. for up to 100 epochs & retain the top 50% of the trials at each checkpoint

Hyperparameter	Search space
<i>Normalization</i>	
Layer Norm	Y, N
Layer Norm Epsilon	loguniform(1e-6, 1e-4)
Batch Norm	Y, N
Grad Value Clipping	Y, N
Clip Grad Value	0.5, 1.0, 5.0
Max Grad Norm	1, 5, 10
Gradient Accumulation Steps	1, 2, 4
<i>Embeddings</i>	
Sinusoidal Embedding (Temporal)	Fixed, Learnable
Split Embeddings	Y, N
Categorical Embedding Dim	32, 64, 128
Numerical Embedding Dim	32, 64, 128
Embedding Weight	0.3, 0.5, 0.7
(Categorical, Numerical, Static, Dynamic)	
Pooling Method	max, mean
<i>Window Size</i>	
Min Window Size	uniform(2, 50)
Max Window Size	uniform(100, 512)
Seq Window Size	uniform(Min Window Size, Max Window Size)

Architecture	
Num Hidden Layers	4, 6, 8
Head Dim	32, 64, 128
Num Attention Heads	4, 8, 12
Intermediate Size	128, 256, 512
Regularization	
Dropout (Input, Intermediate, Residual, Attention)	0.1, 0.2, 0.3
Weight decay	loguniform(1e-5, 1e-2)
Optimization	
Initial LR	loguniform(1e-4, 1e-01)
End LR	loguniform(1e-6, 1e-4)
LR decay	uniform(0.01, 0.5)
Batch Size	128, 256, 512
Patience	1, 5, 10
Use LR Scheduler	Y, N
LR Scheduler Type	cosine, linear, one cycle, reduce on plateau

Hyperparameter Trial Results: Diagnoses & Procedures

(1/2)

Trial	LN	BN	Sin. Emb	Split Emb.	C. Emb. Dim	N. Emb. Dim	C. Emb. Wt	N. Emb. Wt	S. Emb. Wt	D. Emb. Wt
1	Y	Y	Y	Y	32	128	0.5	0.7	0.6	0.5
2	Y	N	N	Y	128	64	0.3	0.5	0.3	0.7
3	Y	Y	N	N	–	–	0.3	0.7	0.4	0.7
4	Y	N	N	N	–	–	0.5	0.5	0.3	0.5
5	N	Y	N	N	–	–	0.3	0.3	0.4	0.5
6	N	N	N	N	–	–	0.5	0.3	0.6	0.5
7	N	N	N	Y	128	32	0.3	0.3	0.4	0.5
8	Y	Y	Y	N	–	–	0.7	0.7	0.3	0.5

Notes: LN=Use Layer Norm; BN=Use Batch Norm; Sin. Emb=Learnable Sinusoidal ATE; Split Emb.=Split Embeddings; C. Emb. Dim=Categorical Embedding Dimension; N. Emb. Dim=Numerical Embedding Dimension; C. Emb. Wt=Categorical Embedding Weight; N. Emb. Wt=Numerical Embedding Weight; S. Emb. Wt=Static Embedding Weight; D. Emb. Wt=Dynamic Embedding Weight.

Hyperparameter Trial Results: Diagnoses & Procedures

(2/2)

Trial	NHL	SWS	HD	NAH	ID	AD	MGN	IS	PM	BS	ILR	GA	P	Val AUC
1	6	168	64	8	0.1	0.1	10	256	mean	1024	0.002	1	5	0.685
2	6	336	64	12	0.3	0.1	1	1024	max	512	0.08	2	5	0.649
3	4	336	32	8	0.1	0.3	5	256	max	1024	1.9×10^{-4}	4	5	0.635
4	4	168	32	4	0.1	0.1	10	1024	max	512	3.4×10^{-4}	2	10	0.636
5	4	336	32	4	0.3	0.3	5	1024	mean	512	7.9×10^{-4}	2	10	0.623
6	8	336	32	4	0.1	0.1	10	512	mean	512	0.002	1	10	0.623
7	8	504	16	8	0.3	0.3	10	1024	max	1024	3.2×10^{-5}	2	5	0.610
8	4	336	16	8	0.1	0.3	10	256	max	1024	2.3×10^{-5}	2	10	0.604

Notes: NHL = Num Hidden Layers; SWS = Sequence Window Size; HD = Head Dimension; NAH = Number of Attention Heads; ID = Intermediate Dropout; AD = Attention Dropout; MGN = Maximum Gradient Norm; IS = Intermediate Size; PM = Pooling Method; BS = Batch Size; ILR = Initial Learning Rate; GA = Gradient Accumulation; P = Patience; Val AUC = Validation AUC.

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